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Original Research

Enhanced recovery after surgery in patients after hip and knee arthroplasty: a systematic review and meta-analysis

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Abstract

Purpose: Enhanced recovery after surgery (ERAS) was characterized as patient-centered, evidence-based, multidisciplinary team-developed routes for a surgical speciality and institution to improve postoperative recovery and attenuate the surgical stress response. However, evidence of their effectiveness in osteoarthroplasty remains sparse. This study aimed to develop an ERAS standard and evaluate the significance of ERAS interventions for postoperative outcomes after primary total hip arthroplasty (THA) or total knee arthroplasty (TKA).

Methods: We searched Medline, Embase, Cochrane databases, and Clinicaltrials.gov for randomized controlled trials, cohort studies, and case–control studies until 24 February 2023. All relevant data were collected from studies meeting the inclusion criteria. Two reviewers independently assessed the risk of bias and extracted data. The primary outcome was the length of stay (LOS), postoperative complications, and readmission rate. The secondary outcomes included transfusion rate, mortality rate, visual analog score (VAS), the Western Ontario and McMaster University Osteoarthritis Index (WOMAC), Short Form 36 (SF-36) bodily pain (SF-36 BP), SF-36 physical function (SF-36 PF), oxford knee score, and range of motion (ROM).

Results: A total of 47 studies involving 76 971 patients (ERAS group: 29 702, control group: 47 269) met the inclusion criteria and were included in the meta-analysis. The result showed that ERAS could significantly shorten the LOS (WMD = −2.65, $P < .001$), reduce transfusion rate (OR = 0.40, $P < .001$), and lower 30-day postoperative mortality (OR = 0.46, $P = .01$) without increasing postoperative complications or readmission rate. Apart from that, ERAS may decrease patients' VAS (WMD = −0.88, $P = .01$) while improving their ROM (WMD = 6.65, $P = .004$), SF-36 BP (WMD = 4.49, $P < .001$), and SF-36 PF (WMD = 3.64, $P < .001$) scores. However, there was no significant difference in WOMAC, oxford knee score between the ERAS and control groups.
Furthermore, we determined that the following seven components of the ERAS program are highly advised: avoid bowel preparation, PONV prophylaxis, standardized anesthesia, use of local anesthetics for infiltration analgesia and nerve blocks, tranexamic acid, prevent hypothermia, and early mobilization.

Conclusion: Our meta-analysis suggested that the ERAS could significantly shorten the LOS, reduce transfusion rate, and lower 30-day postoperative mortality without increasing postoperative complications or readmission rate after THA and TKA. Meanwhile, ERAS could decrease the VAS of patients while improving their ROM, SF-36 BP, and SF-36 PF scores. Finally, we expect future studies to utilize the seven ERAS elements proposed in our meta-analysis to prevent increased readmission rate for patients with THA or TKA.

Key message

What is already known on this topic
A growing number of studies have identified a positive role for ERAS in arthroplasty; however, the clinical effectiveness and elements of ERAS need to be further defined.

What this study adds
This research not only assesses the influence of ERAS on length of stay, complications, and readmission rates, but further evaluates the impact of ERAS on joint function and proposes the recommended seven elements of ERAS.

How this study might affect research, practice, or policy
Given the potential advantages of ERAS, we encourage its use in arthroplasty and anticipate that future research will incorporate our proposed seven ERAS elements to prevent higher readmission rates in THA or TKA patients.

Registration number
CRD42023390678.

Keywords: enhanced recovery after surgery; total hip arthroplasty; total knee arthroplasty; meta-analysis

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Introduction

Enhanced recovery after surgery (ERAS) was advocated by the Danish surgeon Henrik Kehlet in the1990s for accelerated recovery and reduction in complications after major surgeries [1]. Kehlet proposed that the "organ dysfunction" that manifested in pain, nausea, vomiting, exhaustion, and mental instability is caused by surgical stress, metabolic and endocrine abnormalities, as well as extended immobilization. Consequently, reducing the level of organ failure in patients by various means can significantly reduce postoperative problems and enhance recovery. Although a single improvement in preoperative preparation, surgical procedure, anesthetic procedure, or medicine has been reported in many studies, organ dysfunction may not be considerably lessened only by a single improvement.

As for multimode strategy targeting organ dysfunction, ERAS was characterized as patient-centered, evidence-based, multidisciplinary team-developed routes for a surgical speciality and institution to reduce the physiological stress response of patients and speed up their recovery. The cornerstone of ERAS is to use the least intrusive surgical methods available to lower morbidities, shorten hospital stays, save money, boost patient satisfaction, and promote better recovery [2]. Since the debut in colorectal surgery, ERAS has evolved and been extended to cover operations on the liver, pancreas, stomach, breast, and reconstructive surgery. Research has shown that ERAS benefits patients and the healthcare system [3].

With the rise in osteoarthritis patients in recent years, more and more total hip and knee arthroplasty (THA and TKA) are being performed worldwide, usually in aged patients with complex comorbidities. Therefore, many surgeons are concerned about the application effect of ERAS in joint replacement [4]. The main principle of ERAS has also been applied to THA and TKA, leading to improvements in pain management with multimodal opioid-sparing regimens that include a local anesthetic infiltration technique or peripheral nerve blocks, which would facilitate early mobilization and allow functional rehabilitation to begin a few hours after surgery [5]. Catheters, including local drains and urinary catheters, are used as little as possible for early movement after the surgery. In addition, tranexamic acid, antiemetics, antibiotics, antithrombotic medications, bowel preparation avoidance, and preoperative patient education helped patients recover from surgery more rapidly and with less discomfort [6]. The primary goal of ERAS is to improve recovery and decrease morbidity. Secondary benefits, if successful, will include a reduced length of stay (LOS), costs, and recovery [7].

Some researchers have carried out meta-analyses of these trials as more research focuses on the potential of ERAS in osteoarthroplasty. The clinical effectiveness of ERAS, however, is still unknown. Some of these meta-analyses include papers that are not sufficiently comprehensive [8], while others have inadequate outcomes [9]. Additionally, some analyses include an excessive amount of literature without screening [10]. Therefore, we offer a thorough evaluation and meta-analysis of all research reported about the application of ERAS in joint arthroplasty. We anticipate that our study will provide an in-depth examination of all high-quality trials that have been published, which may offer clinical surgeons an evidence-based basis when determining whether to employ ERAS or not.

Methods

This is a systematic review and meta-analysis of previous reports, and we believe that ethical approval is unavailable in our study. Our review followed the guidelines of Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA), and the protocol was registered in PROSPERO (registration number CRD42023390678) before the literature search (Supplementary Table S1: PRISMA Checklist).

Search strategy

Two independent reviewers (Z.Q. and C.Y.) searched Medline, Embase, and Cochrane databases, updated to 24th February 2023, for randomized controlled trials (RCTs), cohort studies, and case–control studies. The search strategy used for the Medline database is available as Supplementary Material (Supplementary Table S2). To expand the search range, the keywords were "enhanced recovery after surgery," "enhanced recovery," "ERAS," "fast-track," "FTS," "accelerated track," "arthroplasty," "replacement," "hip," and "knee." Clinicaltrials.gov was searched for completed but unpublished RCTs. Titles and abstracts were independently reviewed by two researchers (Z.Q. and C.Y.), and then full-text access to any articles that met the inclusion criteria was granted. After that, they independently evaluated full-text articles for eligibility without regard to language. The reference studies of earlier systematic reviews, meta-analyses, and included research were manually searched to prevent initial misses. Any disagreement was discussed and resolved with the third independent reviewers.

Inclusion criteria and exclusion criteria

The inclusion criteria for the studies were as follows: (i) population—human patients undergoing primary hip arthroplasty or knee arthroplasty, with no age, nationality, and race restrictions; (ii) ERAS group—care with ERAS; (iii) control group—standard or traditional care; (iv) at least one of the outcomes—primary outcomes (LOS, complication rate, readmission rate) or secondary outcomes (transfusion rate, mortality rate, etc.); and (v) general information [e.g. gender, age, body mass index (BMI)] of the ERAS group, and the control group was not statistically different at baseline.

The exclusion criteria were as follows: (a) systematic reviews, conference abstracts, case reports, case series, letters, comments, and studies without a control group; (b) duplicate articles; (c) nondetailed data of outcomes or missing data; and (d) human patients who underwent revision, outpatient, or same-day surgical procedure were excluded.

Outcomes of interest

The primary outcomes for this meta-analysis were LOS, complication rate, and readmission rate in the ERAS group and the control group. The secondary results were transfusion rate, mortality rate, visual analog score (VAS), the Western Ontario and McMaster University Osteoarthritis Index (WOMAC), Short Form 36 bodily pain (SF-36 BP), SF-36 physical function (SF-36 PF), oxford knee score (OKS), and range of motion (ROM) in each group.

Data extraction

Two researchers, Z.Q. and C.Y., independently retrieved data from relevant journals. The following study characteristics were extracted: author, journal, publishing year, publishing type, objective, method, conclusion, inclusion criteria, exclusion criteria, ERAS measures, surgical site, and patient characteristics (sample size, gender, age, and BMI), as well as primary and secondary outcomes. To better study the content of ERAS, we additionally analyzed the patient pathways in the ERAS and control groups. If there were any disagreements, the article would be presented to a third author and discussed among the investigators. When information was lacking, the associated author was contacted via email and asked to give more details.

Quality assessment

The quality of the methodology in included studies was independently evaluated by the two authors (Z.Q. and C.Y.). The Cochrane Risk of Bias Assessment Tool (CROBAT) was utilized to assess the quality of included RCTs. CROBAT included "Random sequence generation,", "Allocation concealment," "Blinding of participants and personnel," "Blinding of outcome assessment," "Incomplete outcome data," "Selective reporting," and "Other bias" (Supplementary Table S3.1). There were three options for each question: "Low risk," "Moderate," and "High risk." The risk level of RCTs would be evaluated by researchers based on the information released. The methodological quality of included cohort and case-cohort studies was assessed using the Newcastle–Ottawa Scale (NOS). The NOS comprises eight factors classified into selection, comparability, and outcome (cohort studies) or exposure (case–control studies). For each issue, a variety of response options are provided. The NOS ranges from zero to nine points, with studies receiving six or more points considered excellent quality (Supplementary Tables S3.2 and S3.3). The decision was reached by consulting a third reviewer L.R. in the case of disagreements and failed consensus. Publication bias was evaluated by funnel plots and further confirmed by Egger's test, and $P \leq .05$ was considered the statistically significant risk of bias. We also used the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) tool to evaluate the quality of evidence for the primary outcomes. The GRADE tool classified the evidence of outcomes into "High," "Moderate," "Low," and "Very low." Each assessment could reduce or promote the level of quality. Specific rules are explained in Supplementary Table S3.4.

Statistical analysis

Continuous data using the same scale would be summarized by weighted mean difference (WMD) with 95% confidence intervals (CIs), while using different scales would be measured by standard mean difference with 95% CIs. Dichotomous data would be calculated by odds ratio (OR) with 95% CIs. If the study provided medians and interquartile ranges instead of means and standard deviations (SDs) for continuous variables, medians and interquartile ranges would be converted to means and SDs according to the Cochrane Handbook. All statistical tests were two-tailed, and $P \leq .05$ was regarded as a statistically significant difference. According to Review Manager 5.3, inverse variance (fixed/random-effects model) was used for continuous data and Mantel–Haenszel (fixed/random-effects model) was used for dichotomous data. Heterogeneity in the meta-analysis was assessed using Cochrane Q and I^2 statistics with appropriate analysis models. A fixed-effects model was employed when $I^2 < 50\%$, and a random-effects model was employed when $I^2 \geq 50\%$. Studies with zero events were included by applying the standard continuity corrections (adding 0.5 to all cells for trials with zero events). Subgroup analysis would be carried out when detailed data were available. It would be based on the study type, surgical site, and degree of compliance based on ERAS elements.

GRADE is based on the risk of bias, inconsistency, indirectness, etc., and is rated high when the lower risk criteria are met. See Supplementary Table S3.4 for detailed criteria. Meta-regression analysis was performed on outcomes that included ≥10 studies with high heterogeneity ($I^2 \geq 50\%$). Sensitivity analysis was carried out by excluding each study one at a time to determine whether individual studies determined the efficacy of the outcome. Publication bias for our outcomes was assessed with funnel plots, and the asymmetry of the funnel was evaluated with Egger's test. The trim-and-fill method was also used to assess publication bias for the primary outcomes (when the P-value of Egger's test is <.5). All analyses were completed with Review Manager 5.3 and STATA 16.0.

Results

Search results

Figure 1 depicts a flow diagram of the study selection process. A total of 2462 articles were identified (2442 articles through an electronic database search and 20 identified through other sources). After removing the duplicate studies, 1552 potentially relevant studies were reserved based on the title and abstract. Following that, 90 articles were chosen for full-text review. Finally, 11 RCTs, 27 case–control studies, and 9 cohort studies were included in this meta-analysis.

Study characteristics and quality evaluation

Table 1 provides a summary of the traits of all 47 included research. The papers were all published after 2008 and had similar distributions of sex, age, BMI, and intervention. Although there was a difference in the baseline ages of patients ($P = .005$), we concluded that the difference was insignificant. The 47 studies included 29 702 persons who received ERAS and 47 269 who received a control intervention, and 30 studies reviewed LOS; 17 studies reported the incidence of postoperative complications; 14 studies provided readmission rate; 16 studies mentioned transfusion rate; 10 studies provided mortality rate; 7 studies referred to VAS; 5 studies examined OKS; 3 studies noted WMOAC, SF-36 BP, and SF-36 PF; and 4 studies mentioned ROM.

The quality of the 11 RCTs evaluated using CROBAT is summarized in Supplementary Fig. S1. All of the studies described random sequence generation, and none of the studies had incomplete or selective outcome data. Allocation concealment was described in seven studies. In this type of clinical trial, blinding was challenging to achieve. Supplementary Tables S6 and S7 summarize the quality of nine cohort studies and 27 case–control studies, which were evaluated using the NOS. All cohort and case–control studies with scores ≥6 were considered. Overall, the included studies had high methodological quality.

Enhanced recovery after surgery elements

Table S8 lists the following 17 elements of ERAS protocol for included studies and evaluates the presence or absence of these elements (Supplementary Table S8). Additionally, after carefully examining several excellent earlier reviews [11], we determined that the following seven components of the ERAS program are highly advised: avoid bowel preparation, PONV prophylaxis, standardized anesthesia, use of local anesthetics for infiltration analgesia and nerve blocks, tranexamic acid, prevent hypothermia, and early mobilization. These seven elements comprised 21.3%, 29.8%, 59.6%, 42.6%, 55.3%, 8.5%, and 89.4% of the total. Following discussion, the studies that included at least four of these seven components were considered to be of a high degree of compliance, while the remaining studies were supposed to be of a low degree. The primary outcomes of the article were analyzed based on the degree of compliance with ERAS elements (high- and low-degree subgroups). The percentages of high and low compliance studies were 40.43% and 59.57%, respectively.

Table 1. Characteristics of studies included in systematic review.

Year/author	Study design	Country	Surgical site	Experimental group/control group	Characteristics of study population								Outcome
					Sample size (N)		Gender (N: male/female)[a]		Age (years) mean ± SD[b]		BMI[c]		
					E	C	E	C	E	C	E	C	
2008 Larsen	Case–control study	Denmark	Hip knee	Accelerated intervention group/current intervention group	142	105	68/74	53/52	65 ± 11	65 ± 11	NA	NA	LOS, mortality
2008 Petersen	RCT	Denmark	Hip	Optimized perioperative care/conventional perioperative care	28	33	17/11	17/16	56 (28–84)[d]	58 (26–81)[d]	NA	NA	Transfusion, SF-36 BP, SF-36 PF
2009 Larsen[e]	RCT	Denmark	Hip knee	Accelerated intervention/current intervention	45	42	20/25	23/19	64 ± 10.8	66 ± 9.2	NA	NA	LOS, readmission
2009 Larsen[f]	Case–control study	Denmark	Hip	Accelerated intervention group/current intervention group	55	50	27/28	23/27	65 ± 9.6	67 ± 9.8	NA	NA	LOS
2011 Gulotta	Prospective cohort study	American	Hip	Accelerated postoperative pathway/traditional pathway	149	134	98/51	80/54	50.3 ± 8.9	52.3 ± 8.5	26.9 ± 5.9	27.2 ± 5.8	LOS, complications
2011 Malviya	Case–control study	UK	Hip knee	"Enhanced recovery" protocol/traditional recovery	1500	3000	711/789	1482/1518	68[g]	69[g]	NA	NA	Transfusion, mortality
2012 den Hertog	RCT	Germany	Knee	Fast-track rehabilitation/standard postoperative rehabilitation care	74	73	23/51	20/53	66.58 ± 8.21	68.25 ± 7.91	31.17 ± 5.82	30.38 ± 6.05	LOS, complications, WOMAC
2012 Gooch	RCT	Canada	Hip knee	Evidence-informed accelerated clinical pathway/standard of care	1066	504	422/644	202/302	69.0 ± 11.1	69.0 ± 10.4	29.5 ± 5.6	29.4 ± 5.4	Complications, mortality, WOMAC, SF-36 BP, SF-36 PF
2012 McDonald	Case–control study	UK	Knee	ERP/control group	1081	735	439/642	307/428	69[d]	70[d]	31[d]	31d	Transfusion, mortality, OKS
2013 Scott	Case–control study	UK	Hip knee	ERAS/non-ERAS	687	1936	NA	NA	68 ± 11	68 ± 10	NA	NA	Transfusion
2014 Glassou	Retrospective cohort study	Denmark	Hip knee	Fast-track program/national comparison cohort	5740	19 578	2411/3329	8027/11551	68 ± 10	69 ± 10	NA	NA	Complications, mortality
2014 Khan	Case–control study	UK	Hip Knee	ER/traditional recovery	3000	3000	1390/1610	1482/1518	68 ± 10	69 ± 10	NA	NA	Mortality
2015 Auyong	Case–control study	American	Knee	Post ERAS pathway/pre ERAS pathway	126	126	41/85	44/82	66.02 ± 10.02	68.44 ± 9.98	31.88 ± 7.629	31.3 ± 6.562	Readmission, transfusion
2015 Christelis	Case–control study	Australia	Hip knee	ERAS/pre-ERAS (existing-practice)	297	412	113/299	164/133	67 ± 10	68 ± 11	No statistically significant		LOS, complications, readmission, transfusion , mortality, VAS
2015 Köksal	Case–control study	Turkey	Knee	Rapid recovery protocol/standard protocol	84	85	38/46	40/45	64 (56–79)[d]	68 (55–77)[d]	31.6 ± 6.2	31.2 ± 7.1	LOS, ROM

(*Continued*)

Table 1. Continued

Year/author	Study design	Country	Surgical site	Experimental group/control group	Characteristics of study population								Outcome
					Sample size (N)		Gender (N: male/female)[a]		Age (years) mean ± SD[b]		BMI[c]		
					E	C	E	C	E	C	E	C	
2015 Maempel	Case–control study	UK	Knee	ERP/non-ERP group	84	81	42/42	37/44	69.8 ± 8.9	70.1 ± 10.5	32 (22.6–46.6)[d]	31.8 (20.5–41.9)[d]	Transfusion
2016 Stowers	Case–control study	New Zealand	Hip knee	ERAS/non-ERAS	100	100	47/53	41/59	66.7 ± 9.2	65.4 ± 12.5	34.4 ± 7.0	32.5 ± 6.8	Complications, readmission
2016 Talboys	Retrospective cohort study	UK	Hip	ERP/routine preoperative and postoperative care	50	50	12/38	15/35	83 ± 7.8	85 ± 8.2	NA	NA	LOS, mortality
2016 Yang	RCT	China	Hip	Fast-track program/standard program group	123	127	56/70	65/67	64.2 ± 9.4	66.3 ± 8.6	25.2 ± 4.2	25.3 ± 5.0	LOS, complications, readmission, transfusion
2017 Castorina	Retrospective cohort study	Italy	Knee	Fast-track/traditional methods	95	37	NA	NA	71.1 ± 7.77	74.62 ± 6.42	NA	NA	ROM, transfusion
2017 Dwyer	Case–control study	UK	Knee	ERP/non-ERP	57	55	17/40	22/33	70 ± 9	73 ± 7	NA	NA	LOS
2017 Galbraith	Case–control study	Ireland	Hip knee	ERP/non-ERP	Total:165 Hip:125 Knee:40	Total:145 Hip:119 Knee:26	95/70	75/70	NA	NA	NA	NA	LOS, readmission
2017 Gwynne-Jones	Case–control study	New Zealand	Hip knee	ERAS/pre-ERAS	Total:528 Hip:318 Knee:210	Total:507 Hip:314 Knee:193	Total: 253/275 Hip:146/172 Knee: 107/103	Total: 229/278 Hip:146/168 Knee: 83/110	Total: 69.1352 ± 10.7809 Hip:68.3 ± 11.8 Knee: 70.4 ± 8.9	Total: 67.942 ± 10.9083 Hip:66.8 ± 11.8 Knee: 69.8 ± 9.0	NA	NA	LOS, readmission, mortality, OKS
2017 Wilches	Case–control study	Spain	Hip knee	Fast-track/control group	100	100	40/60	40/60	69.24 ± 9.64	73.07 ± 8.33	NA	NA	LOS, complications, transfusion
2018 Berg	Case–control study	Sweden	Hip knee	Fast-track/control group	Hip: 3857 Knee:3413	Hip: 3724 Knee:2916	Hip: 1657/2200 Knee: 1461/1952	Hip: 1538/2186 Knee: 1182/1734	Hip: 69.5 ± 10.1 Knee: 68.8 ± 9.0	Hip: 69.5 ± 10.3 Knee: 69.5 ± 9.3	Hip: 27.4 ± 4.6 Knee: 29.2 ± 4.60	Hip: 27.6 ± 4.6 Knee: 29.2 ± 4.9	Complications, Readmission
2018 Featherall	Case–control study	American	Hip	Evidence-based care pathway/preprotocol or historical control cohort	2081	2000	1033/1048	960/1040	64.09 ± 12.04	64.03 ± 12.09	30.13 ± 6.17	30.09 ± 6.38	LOS, complications
2018 Fransen	RCT	Netherlands	Knee	Fast-track protocols/regular joint care protocol	25	24	11/14	9/15	64 ± 9	61 ± 7	28.7 ± 3.5	30.0 ± 4.1	LOS, complications, VAS
2018 Pamilo[e]	Case—control study	Finland	Knee	Fast-tracking/traditional recovery	624	437	No statistically significant		No statistically significant		NA	NA	Readmission
2018 Pamilo[f]	Case–control study	Finland	Hip	Fast-tracking/traditional recovery	437	464	No statistically significant		No statistically significant		NA	NA	Readmission
2018 Yanik	Case–control study	American	Hip knee	Rapid recovery protocol/control group	78	174	70/8	157/17	66.56 ± 8.63	65.89 ± 8.70	30.95 ± 4.79	32.73 ± 5.77	LOS
2019 Didden	Case–control study	Netherlands	Knee	nNew pathway/old pathway	85	85	40/45	36/49	69.0 (52–86)[d]	69.0 (47–86)[d]	28.7 (20.9–42.6)[d]	28.4 (21.8–43.2)[d]	Readmission

(*Continued*)

Table 1. Continued

Year/author	Study design	Country	Surgical site	Experimental group/control group	Characteristics of study population: Sample size (N) E	Sample size (N) C	Gender (N: male/female)[a] E	Gender C	Age (years) mean ± SD[b] E	Age C	BMI[c] E	BMI C	Outcome
2019 Jiang	Prospective cohort study	China	Knee	ERAS group/traditional group	106	141	58/48	83/58	74.2 ± 6.3	75.4 ± 5.9	32.1 ± 5.1	31.4 ± 4.5	LOS, complications, transfusion, VAS, ROM
2020 Jansen	Retrospective cohort study	Netherlands	Knee	Fast-track pathway/nonfast-track pathway	403	283	280/123	205/78	67.46 ± 8.67	68.31 ± 9.37	28.89 ± 4.5	29.52 ± 4.7	OKS
2020 Plessl	Case–control study	American	Knee	Rapid recovery protocols/standard recovery protocol	194	129	59/135	47/82	68.0 ± 8.8	65.7 ± 9.8	31.2 ± 5.1	32.7 ± 5.9	LOS
2020 Ripollés-Melchor	Prospective cohort study	Spain	Hip knee	ERAS group/no ERAS group	1592	4554	662/930	1904/2650	70 (63–76)[d]	71 (64–76)[d]	Median (IQR) 29.3 (26.4–32.9)	29.4 (26.4–32.8)	Complications, readmission, transfusion, mortality,
2020 Zhang	RCT	China	Hip	clinical nursing pathway/routine nursing	35	35	14/21	15/20	58.77 ± 13.97	59.66 ± 11.24	NA	NA	LOS, SF-36 BP, SF-36 PF
2021 Arienti	Prospective cohort study	Italy	Knee	Fast-track rehabilitation protocol/conventional rehabilitation protocol	20	23	6/14	4/19	69 (60–77)[d]	69 (65–73)[d]	Median (IQR) 27.7 (25.4–29.4)	29.4 (26.2–32.5)	LOS
2021 Chung	Case–control study	Hongkong	Hip knee	ERAS/conventional practices	111	117	44/67	39/78	69.6 ± 9.2	67.9 ± 12.4	NA	NA	LOS
2021 de Carvalho	Case–control study	Brazil	Hip	Fast-track/control group	47	51	25/22	25/26	No statistically significant		NA	NA	LOS, complications, transfusion
2021 Gleicher	Case–control study	Canada	Knee	ER/preintervention	383	232	155/228	84/148	66.5 ± 9.9	66.1 ± 10.1	32.45 ± 7.83	31.0 ± 7.0	LOS
2021 Picart	Case–control study	France	Knee	Fast-track group/Conventional group	216	335	78/138	114/221	69.23 ± 7.80	69.41 ± 9.86	30.15 ± 4.79	30.86 ± 5.46	LOS, readmission, transfusion
2021 Wei	RCT	China	Knee	ERAS/Control group	35	34	7/28	7/27	65.77 ± 4.51	65.62 ± 5.06	26.66 ± 1.19	26.91 ± 1.22	LOS, VAS
2021 Wu	RCT	China	Hip	ERAS/routine nursing	16	16	NA	NA	35.6 ± 6.7	31.7 ± 9.2	NA	NA	LOS, complications, VAS
2021 Zhong	Prospective cohort study	China	Hip	Rapid rehabilitation surgery group/conventional rehabilitation group	180	168	78/102	73/95	65 ± 11.2	64 ± 12.1	23.8 ± 3.5	24.1 ± 3.6	LOS, VAS
2022 Aguado-Maestro	RCT	Spain	Knee	Rapid recovery program/classic protocol	83	92	32/51	35/57	70.7[g]	71.3[g]	30.1	30.6	LOS, complications, ROM, OKS
2022 Azam	Case–control study	India	Knee	Enhanced recovery/standard treatment group	275	190	77/198	65/125	65.8 ± 9.2	66.1 ± 10.1	24.6 ± 4.5	25.1 ± 3.9	LOS, transfusion, VAS, OKS
2022 Elmoghazy	RCT	Germany	Hip	Fast-track rehabilitation/conventional rehabilitation	30	30	20/10	10/20	65 ± 9	72 ± 7	28 ± 5	28.4 ± 5	LOS, complications, readmission, transfusion, WOMAC

[a]Comparison of gender baseline between experimental group and control group: OR = .95, 95%CI (0.84, 1.0), P = .434. [b]Comparison of age baseline between experimental group and control group: WMD = −.53, 95%CI (−0.90, −0.16), P = .005. [c]Comparison of BMI baseline between experimental group and control group: WMD = −0.15, 95%CI (−0.37, −0.06), P = .167. [d]The values are given as the median, with or without the range in parentheses. [e]These are two different articles. [f]These are two different articles. [g]The values are given as the mean, without the SD. No statistically significant: There is no significant difference in the article. ERP: enhanced recovery program; ER: enhanced recovery; C: control group; E: experimental group (ERAS); N: number of patients; NA: not available; .

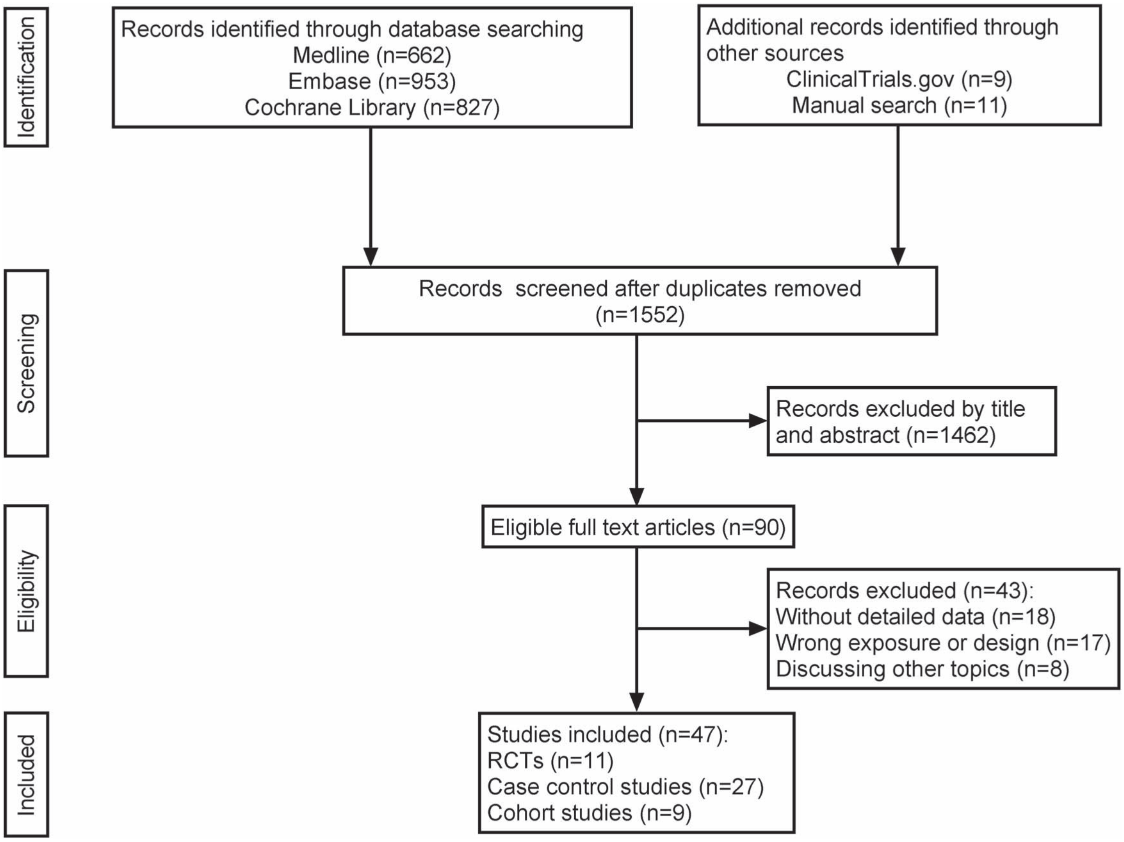

Figure 1 Flow diagram of the study selection process

Primary outcomes

Effect of enhanced recovery after surgery protocols on hospital length of stay

Thirty studies [12–40, 46] reported on the LOS. A total of 11 460 patients were evaluated for quantitative analysis, with 5784 in the ERAS group and 5676 in the control group (Fig. 2). The result showed that LOS in the ERAS group was considerably lower than that in the control group [WMD = −2.65 (−3.03, −2.26), $P < .001$]. Furthermore, in comparison to the control group, ERAS was associated with a statistically significant decrease in the LOS in different study types and surgical sites: RCTs [WMD = −3.08 (−3.95, −2.21), $P < .001$], nonrandomized controlled trials [NRCT; WMD = −2.45 (−2.85, −2.04), $P < .001$], THA [WMD = −2.60 (−3.75, −1.45), $P < .001$], TKA [WMD = −2.84 (−3.59, −2.08), $P < .001$], and THA + TKA [WMD = −2.38 (−3.23, −1.53), $P < .001$] (Table 2). The research considered to be of a high degree of compliance in terms of ERAS protocol elements also demonstrated substantially lower LOS in the ERAS group than in the control group [WMD = −2.29 (−3.20, −1.38), $P < .001$], though the same results were observed in the low-degree research [WMD = −2.87 (−3.35, −2.39), $P < .001$] (Table 2). However, there were signs of statistical heterogeneity between studies ($I^2 > 95\%$). Meta-regression analysis could not find causes of heterogeneity according to the types of study design, surgical sites, and the degree of compliance of ERAS. Sensitivity analysis found that none of the studies impacted the outcome (Supplementary Fig. S2.1). The GRADE table shows low evidence for LOS (Supplementary Table S5).

Effect of enhanced recovery after surgery on postoperative complications

There are 17 papers [15–17, 20, 24–26, 28, 33, 37, 39, 41–46] providing information on postoperative complications. According to sensitivity analysis, a study [15] had a considerable impact on the pooled OR, and its removal could lead to a stable outcome with significantly less statistical heterogeneity (Supplementary Fig. S2.2). Seventeen data from 16 papers were used in the meta-analysis. A total of 53 192 patients were evaluated for quantitative analysis, with 18 750 in the ERAS group and 34 442 in the control group (Fig. 3). There was no difference in the incidence of postoperative complications between the ERAS group and the control group [OR = 0.97 (0.90, 1.04), $P = .374$]. However, in the subgroup of RCTs, ERAS had a considerably lower risk of postoperative complications than the control group [OR = 0.68 (0.48, 0.95), $P = .023$], though there was no significant difference in NRCTs [OR = 0.98 (0.91, 1.06), $P = .68$] (Table 2). Additionally, there was no difference in the incidence of postoperative complications between the ERAS and control group in all surgical sites ($P > .05$) (Table 2). Our findings are unaffected by the degree of compliance in ERAS elements as well, as the pooled OR for high-degree literature is 0.97 [(0.89, 1.05), $P = .42$], while the pooled OR for low-degree literature is 0.97 [(0.82, 1.14), $P = .71$] (Table 2). The statistical heterogeneity among studies was low ($I^2 < 50\%$). The GRADE of the NRCTs was very low evidence, whereas the GRADE of the RCTs for the incidence of complications was moderate evidence (Supplementary Table S5).

Effect of enhanced recovery after surgery on readmission rate

Postoperative readmission rates were provided in 14 studies [13, 17, 20, 22, 23, 35, 43–49], and a total of 20 datasets were included in the meta-analysis. The readmission rate for a total of 41 728 patients was assessed, with 20 090 in the ERAS group and 21 638 in the control group (Fig. 4). According to the overall meta-analysis, there was no discernible difference between the two groups

Study or Subgroup	Experimental Mean	Experimental SD	Experimental Total	Control Mean	Control SD	Control Total	Weight	Mean Difference IV, Random, 95% CI
1.2.1 RCT								
2021 Wei	3.11	0.32	35	7.06	0.6	34	3.7%	-3.95 [-4.18, -3.72]
2016 Yang	2.1	0.5	123	5.82	1.8	127	3.7%	-3.72 [-4.05, -3.39]
2022 Aguado-Maestro	2.54	0.95	83	3.84	2.27	92	3.6%	-1.30 [-1.81, -0.79]
2022 Elmoghazy	4.5	0.6	30	7.8	1.7	30	3.4%	-3.30 [-3.95, -2.65]
2021 Wu	14.2	0.8	16	16.6	1.3	16	3.3%	-2.40 [-3.15, -1.65]
2012 den Hertog	6.75	2.92	74	13.2	1.66	73	3.3%	-6.45 [-7.22, -5.68]
2018 Fransen	3.7	1.8	25	4.7	1.3	24	3.2%	-1.00 [-1.88, -0.12]
2009 Larsen Ⅰ	4.9	2.4	45	7.8	2.1	42	3.1%	-2.90 [-3.85, -1.95]
2020 Zhang	16.31	3.46	35	18.69	4.78	35	1.9%	-2.38 [-4.33, -0.43]
Subtotal (95% CI)			**466**			**473**	**29.3%**	**-3.08 [-3.95, -2.21]**
Heterogeneity: Tau² = 1.61; Chi² = 188.70, df = 8 (P < 0.00001); I² = 96%								
Test for overall effect: Z = 6.91 (P < 0.00001)								
1.2.2 NRCT								
2020 Plessl	0.8	0.1	194	2.5	0.1	129	3.8%	-1.70 [-1.72, -1.68]
2018 Featherall	2.55	1.46	2081	3.21	1.5	2000	3.8%	-0.66 [-0.75, -0.57]
2021 Gleicher	2.13	1.09	383	2.82	1.25	232	3.8%	-0.69 [-0.88, -0.50]
2017 Gwynne-Jones	4.5	1.88	528	5.64	1.99	507	3.7%	-1.14 [-1.38, -0.90]
2015 Christelis	4.9	1.6	297	5.3	1.6	412	3.7%	-0.40 [-0.64, -0.16]
2021 Picart	3.17	1.59	216	7.25	1.85	335	3.7%	-4.08 [-4.37, -3.79]
2011 Gulotta	2.6	0.9	149	4.1	1.5	134	3.7%	-1.50 [-1.79, -1.21]
2018 Yanik	1.68	1.27	78	3.19	1.35	174	3.7%	-1.51 [-1.86, -1.16]
2021 Zhong	11.5	1.2	180	15.5	2.3	168	3.7%	-4.00 [-4.39, -3.61]
2021 Chung	3.28	1.04	111	5.16	2.06	117	3.6%	-1.88 [-2.30, -1.46]
2019 Jiang	9.6	1.6	106	11.3	1.9	141	3.6%	-1.70 [-2.14, -1.26]
2021 de Carvalho	2.3	0.8	47	6.4	1.5	51	3.6%	-4.10 [-4.57, -3.63]
2022 Azam	3.9	2.1	275	7.5	3.2	190	3.6%	-3.60 [-4.12, -3.08]
2015 Köksal	3.7	1.3	84	6.3	2.5	85	3.5%	-2.60 [-3.20, -2.00]
2008 Larsen	4.3	1.8	142	8.8	3	105	3.4%	-4.50 [-5.15, -3.85]
2017 Galbraith	5.1	2.79	165	8.79	3.31	145	3.4%	-3.69 [-4.38, -3.00]
2009 Larsen Ⅱ	4.2	1.8	55	7.3	1.8	50	3.4%	-3.10 [-3.79, -2.41]
2017 Wilches	4.82	1.7	100	8.15	3.1	100	3.4%	-3.33 [-4.02, -2.64]
2017 Dwyer	5.92	2.25	57	7.7	2.71	55	3.1%	-1.78 [-2.70, -0.86]
2021 Arienti	10.6	3.2	20	17.3	3.8	23	1.8%	-6.70 [-8.79, -4.61]
2016 Talboys	8.5	9.4	50	7	11.2	50	0.7%	1.50 [-2.55, 5.55]
Subtotal (95% CI)			**5318**			**5203**	**70.7%**	**-2.45 [-2.85, -2.04]**
Heterogeneity: Tau² = 0.79; Chi² = 1445.39, df = 20 (P < 0.00001); I² = 99%								
Test for overall effect: Z = 11.79 (P < 0.00001)								
Total (95% CI)			**5784**			**5676**	**100.0%**	**-2.65 [-3.03, -2.26]**
Heterogeneity: Tau² = 1.02; Chi² = 2171.58, df = 29 (P < 0.00001); I² = 99%								
Test for overall effect: Z = 13.42 (P < 0.00001)								
Test for subgroup differences: Chi² = 1.66, df = 1 (P = 0.20), I² = 39.8%								

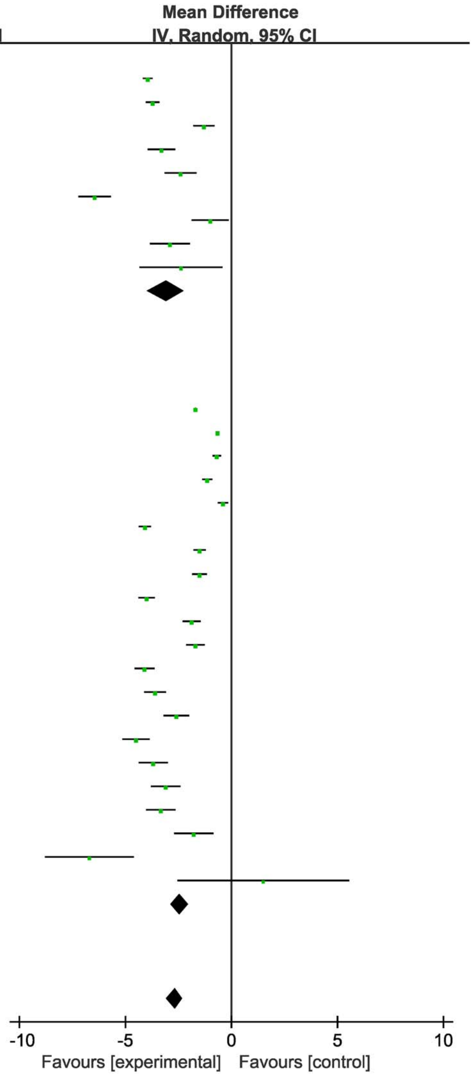

Figure 2 Comparison of LOS between the ERAS and the control groups

[OR = 1.06 (0.98, 1.16), $P = .15$]. Moreover, ERAS did not affect the readmission rate, according to a subgroup analysis based on time, study type, and surgical site (Table 2). Similarly, in the high-degree literature, the readmission rate in the ERAS group was the same as in the control group [OR = 1.03 (0.94, 1.13), $P = .490$]. However, in the low-degree literature, the readmission rate in the ERAS group was much higher than in the control group [OR = 1.37 (1.05, 1.77), $P = .019$] (Table 2). The statistical heterogeneity among studies was low ($I^2 < 50\%$). According to the sensitivity analysis, none of the studies influenced the pooled OR (Supplementary Fig. S2.3). The readmission rate for RCTs was moderate evidence, compared with very low GRADE for NRCTs (Supplementary Table S5).

Secondary outcomes

Effect of enhanced recovery after surgery on transfusion rate

Data on the transfusion rate were provided in 16 trials [17, 20, 24, 28, 33, 35, 40, 45–47, 50–55]. A total of 18 275 patients were involved, including 6387 in the ERAS group and 11 888 in the control group. According to the overall meta-analysis, there was a substantially fewer transfusion rate in the ERAS group than in the control group [OR = 0.40 (0.29, 0.54), $P < .001$] (Table 2). Furthermore, the ERAS group also had a significantly lower transfusion rate than the control group in the subgroups of RCTs, NRCTs, TKA, and THA + TKA. However, the analysis of the THA subgroups failed to achieve statistical significance (Table 2). Meta-regression analysis identified heterogeneity from surgical sites ($P = 0.042$), confirming the need for subgroup analysis. The results of the meta-analysis and heterogeneity are detailed in Table 2. According to the sensitivity analysis, none of the studies had a bearing on the pooled OR (Supplementary Fig. S2.).

Effect of enhanced recovery after surgery on mortality rate

Ten researchers reported postoperative mortality, and 13 datasets from 10 studies [12, 17, 19, 23, 41, 42, 45, 51, 52, 56] were considered in the meta-analysis. A total of 59 336 patients had their death rates assessed; 20 384 were in the ERAS group and 38 952 were in the control group. Between the ERAS and the control groups, there was no difference in the postoperative mortality [OR = 0.76 (0.57, 1.01), $P = .056$] (Table 2). However, ERAS had significantly lower postoperative mortality than the control group at 30 days [OR = 0.46 (0.25, 0.85), $P = .01$], though there was no significant difference at 30–90 days [OR = 0.88 (0.64, 1.21), $P = .43$] (Table 2). The results of heterogeneity are detailed in Table 2. According to the sensitivity analysis, none of the studies impacted the pooled OR (Supplementary Fig. S2.).

Table 2. Results for meta-analysis.

Outcomes	Subgroup analyses	No. of data	Sample (E/C)	OR/WMD	95%CI	P-value	Heterogeneity (I^2 and P)	Egger's test (P-value)
LOS	Overall	30	5784/5676	−2.65	[−3.03, −2.26]	<.001	I^2 = 98.7% P < 0.001	.066
	Study type							
	RCT	9	466/473	−3.08	[−3.95, −2.21]	<.001	I^2 = 95.8% P < .001	.347
	NRCT	21	5318/5203	−2.45	[−2.85, −2.04]	<.001	I^2 = 98.6% P < .001	.356
	Surgery site							
	Hip	10	2766/2661	−2.60	[−3.75, −1.45]	<.001	I^2 = 98.9% P < .001	.028
	Knee	12	1552/1413	−2.84	[−3.59, −2.08]	<.001	I^2 = 98.9% P < .001	.125
	Hip+knee	8	1466/1602	−2.38	[−3.23, −1.53]	<.001	I^2 = 97.2% P < .001	.004
	grade							
	High degree	11	1731/1791	−2.29	[−3.20, −1.38]	<.001	I^2 = 98.5% P < .001	.77
	Low degree	19	4053/3885	−2.87	[−3.35, −2.39]	<.001	I^2 = 98.5% P < .001	.15
Complication rate	Overall	17	18 750/34 442	0.97	[0.90, 1.04]	.374	I^2 = 29.3% P = .124	.002
	Study type							
	RCT	7	1417/866	0.68	[0.48, 0.95]	.023	I^2 = 10.7% P = .348	.021
	NRCT	10	17 333/33 576	0.98	[0.91, 1.06]	.68	I^2 = 24.9% P = .214	.145
	Surgery site							
	Hip	6	6154/5948	0.99	[0.88, 1.13]	.93	I^2 = 47.0% P < .001	.325
	Knee	5	3701/3246	0.93	[0.79, 1.10]	.40	I^2 = 43.1% P = .134	.036
	Hip+knee	6	8895/25 248	0.96	[0.87, 1.07]	.49	I^2 = 13.8% P = .326	.064
	Grade							
	High degree	10	15 327/31 660	0.97	[0.89, 1.05]	.42	I^2 = 46.1% P = .054	.0
	Low degree	7	3423/2782	0.97	[0.82, 1.14]	.71	I^2 = 0% P = .43	.296
Readmission rate	Overall	20	20 090/21 638	1.06	[0.98, 1.16]	.15	I^2 = 40.4% P = .032	.485
	Time/day							
	≤30 days	9	10 820/12 947	1.13	[0.89, 1.44]	.32	I^2 = 50.4% P = .041	.663
	30–90 days	11	9270/8691	1.06	[0.95, 1.19]	.31	I^2 = 36.4% P = .108	.634
	Study type							
	RCT	3	198/199	0.80	[0.43, 1.49]	.488	I^2 = 0.0% P = .016	.954
	NRCT	17	19 892/21 439	1.07	[0.98, 1.17]	.125	I^2 = 47.2% P = .673	.337
	Surgery site							
	Hip	8	8991/8771	1.34	[0.99, 1.81]	.147	I^2 = 64.3% P = .006	.164
	Knee	7	8537/7252	1.00	[0.87, 1.13]	.931	I^2 = 0.0% P = .854	.036
	Hip+knee	5	2562/5615	1.25	[0.95, 1.65]	.109	I^2 = 42.4% P = .139	.568
	Grade							
	High degree	9	17 206/19 006	1.03	[0.94, 1.13]	.490	I^2 = 17.5% P = .287	.959
	Low degree	11	2884/2632	1.37	[1.05, 1.77]	.019	I^2 = 47.3% P = .041	.665

(Continued)

Table 2. Continued

Outcomes	Subgroup analyses	No. of data	Sample (E/C)	OR/WMD	95%CI	P-value	Heterogeneity (I^2 and P)	Egger's test (P-value)
Transfusion rate	Overall	16	6387/11 888	0.40	[0.29, 0.54]	<.001	I^2 = 64.4% P < .001	.682
	Study type							
	RCT	3	181/190	0.47	[0.24, 0.94]	.03	I^2 = 17.5% P = .298	.679
	NRCT	13	6206/11 698	0.38	[0.27, 0.53]	<.001	I^2 = 69.6% P < .001	.498
	Surgery site							
	Hip	4	228/241	0.23	[0.03, 1.74]	.16	I^2 = 83.9% P < .001	.875
	Knee	7	1983/1645	0.37	[0.22, 0.63]	<.001	I^2 = 58.8% P = .024	.208
	Hip+knee	5	4176/10 002	0.47	[0.35, 0.63]	<.001	I^2 = 47.0% P = .110	.205
Mortality rate	Overall	13	20 384/38 952	0.76	[0.57, 1.01]	.056	I^2 = 45.0% P = .040	.441
	Time/day							
	≤30 days	6	8096/11 615	0.46	[0.25, 0.85]	.01	I^2 = 30.3% P = .208	.329
	30–90 days	7	12 288/27 337	0.88	[0.64, 1.21]	.43	I^2 = 47.0% P = .079	.522
VAS	Overall	7	934/985	−0.88	[−1.57, −0.20]	.01	I^2 = 98.6% P < .001	.084
	Study type							
	RCT	3	76/74	−0.56	[−1.16, −0.03]	.06	I^2 = 96.2% P < .001	.219
	NRCT	4	858/911	−1.10	[−2.19, −0.02]	.046	I^2 = 98.1% P < .001	.414
WOMAC	Overall	3	1170/607	−2.86	[−10.47, 4.75]	.46	I^2 = 88.3% P < .001	.851
SF-36 BP	Overall	3	1129/572	4.49	[2.69, 6.29]	<.001	I^2 = 0.0% P = .517	.54
SF-36 PF	Overall	3	1129/572	3.64	[1.79, 5.48]	<.001	I^2 = 0.0% P = .383	.58
OKS	Overall	5	2370/1807	0.33	[−0.93, 1.60]	.61	I^2 = 97.8% P < .001	.418
ROM	Overall	4	380/378	6.65	[2.09, 11.21]	.004	I^2 = 93.3% P < .001	.929

Effect of enhanced recovery after surgery on visual analog score

The pain levels of patients were scored utilizing the VAS, and the score was inversely correlated with their level of discomfort. Seven studies [17, 26, 28, 36–38, 40] reported the VAS within 3 days postsurgery. All investigations showed no significant differences in the baseline VAS scores (P > .05) (Supplementary Table S9). A total of 1919 patients were assessed for VAS, with 934 in the ERAS group and 985 in the control group. Patients in the ERAS group had a considerably lower VAS score than those in the control group [WMD = −0.88 (−1.57, −0.20), P = .01] (Table 2). This was also true in the subgroup of NRCTs [WMD = −1.10 (−2.19, −0.02), P = .046], while there was no significant difference in RCTs [WMD = −0.56 (−1.16, −0.03), P = .06] (Table 2). The results of heterogeneity are detailed in Table 2. According to the sensitivity analysis, none of the studies impacted the pooled WMD (Supplementary Fig. S2.6).

Effect of enhanced recovery after surgery on Western Ontario and McMaster University Osteoarthritis Index score

WOMAC score is divided into three subscales: pain, stiffness, and physical function, with a higher score indicating a more severe condition. Three studies [16, 41, 46] provided these scores over 3 months postsurgery with no significant differences in baseline WOMAC score levels (P > .05) (Supplementary Table S9). A total of 1777 patients were evaluated for WOMAC scores, with 1170 in the ERAS group and 607 in the control group. The pooled WMD showed that WOMAC score was not statistically significantly lower in either group due to ERAS (Table 2).

Effect of enhanced recovery after surgery on short form 36 bodily pain and short form 36 physical function scores

The SF-36 is a generic, self-administered instrument for measuring different aspects of quality of life, with a higher score indicating better health status. Three studies [30, 41, 50] reported PF and BP scores of the SF-36 over 3 months postsurgery, and there were no significant differences in baseline scores levels across all studies (P > .05) (Supplementary Table S9). A total of 1701 patients were evaluated for SF-36 BP and SF-36 PF, with 1129 in the ERAS group and 572 in the control group. The pooled results of both SF-36 BP and PF showed that the ERAS group had a significantly higher score than the control group [SF-36 BP: WMD = 4.49 (2.69, 6.29), P < .001. SF-36 PF: WMD = 3.64 (1.79, 5.48), P < .001]. They all had very low heterogeneity (Table 2).

Study or Subgroup	ERAS Events	ERAS Total	Control Events	Control Total	Weight	Odds Ratio M-H, Fixed, 95% CI
2.3.1 Hip						
2018 Berg (hip)	278	3857	265	3724	17.1%	1.01 [0.85, 1.21]
2018 Featherall	250	2081	236	2000	14.5%	1.02 [0.84, 1.23]
2016 Yang	7	123	13	127	0.8%	0.53 [0.20, 1.37]
2021 Wu	3	16	10	16	0.6%	0.14 [0.03, 0.69]
2022 Elmoghazy	1	30	2	30	0.1%	0.48 [0.04, 5.63]
2021 de Carvalho	5	47	2	51	0.1%	2.92 [0.54, 15.82]
2011 Gulotta	63	149	100	134		Not estimable
Subtotal (95% CI)		**6154**		**5948**	**33.3%**	**0.99 [0.88, 1.13]**
Total events	544		528			
Heterogeneity: Chi² = 9.43, df = 5 (P = 0.09); I² = 47%						
Test for overall effect: Z = 0.08 (P = 0.93)						
2.3.2 Knee						
2018 Berg(knee)	275	3413	229	2916	15.6%	1.03 [0.86, 1.23]
2019 Jiang	26	106	52	141	2.3%	0.56 [0.32, 0.97]
2022 Aguado-Maestro	8	83	13	92	0.8%	0.65 [0.25, 1.65]
2012 den Hertog	8	74	12	73	0.7%	0.62 [0.24, 1.61]
2018 Fransen	2	25	5	24	0.3%	0.33 [0.06, 1.90]
Subtotal (95% CI)		**3701**		**3246**	**19.7%**	**0.93 [0.79, 1.10]**
Total events	319		311			
Heterogeneity: Chi² = 7.03, df = 4 (P = 0.13); I² = 43%						
Test for overall effect: Z = 0.84 (P = 0.40)						
2.3.3 Hip+Knee						
2014 Glassou	246	5740	791	19578	23.5%	1.06 [0.92, 1.23]
2020 Ripollés-Melchor	163	1592	517	4554	16.5%	0.89 [0.74, 1.07]
2015 Christelis	31	297	51	412	2.6%	0.82 [0.51, 1.32]
2012 Gooch	51	1066	25	504	2.2%	0.96 [0.59, 1.57]
2016 Stowers	15	100	22	100	1.3%	0.63 [0.30, 1.29]
2017 Wilches	8	100	14	100	0.9%	0.53 [0.21, 1.34]
Subtotal (95% CI)		**8895**		**25248**	**47.0%**	**0.96 [0.87, 1.07]**
Total events	514		1420			
Heterogeneity: Chi² = 5.80, df = 5 (P = 0.33); I² = 14%						
Test for overall effect: Z = 0.69 (P = 0.49)						
Total (95% CI)		**18750**		**34442**	**100.0%**	**0.97 [0.90, 1.04]**
Total events	1377		2259			
Heterogeneity: Chi² = 22.64, df = 16 (P = 0.12); I² = 29%						
Test for overall effect: Z = 0.89 (P = 0.37)						
Test for subgroup differences: Chi² = 0.40, df = 2 (P = 0.82), I² = 0%						

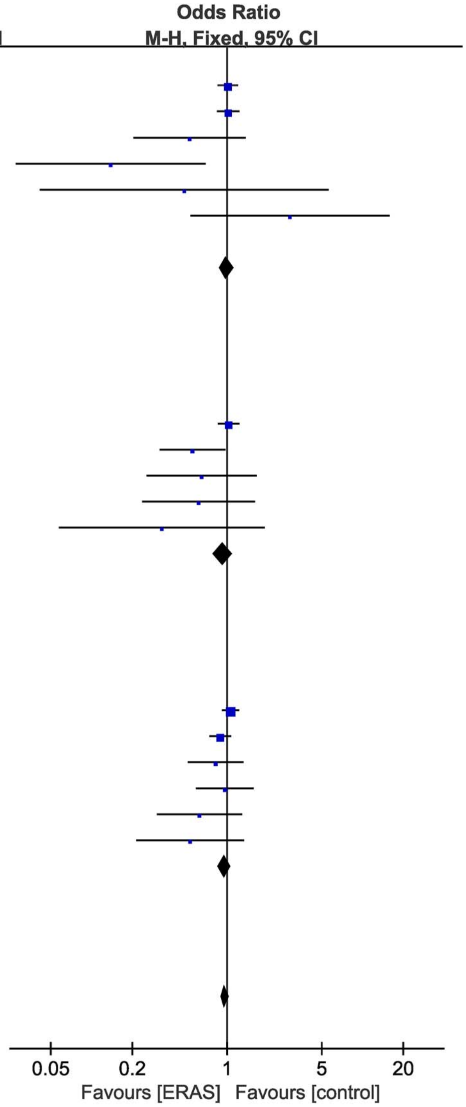

Figure 3 Comparison of complication rate between the ERAS and control groups

Effect of enhanced recovery after surgery on oxford knee score

Featuring a higher score indicating better health status, the OKS was designed exclusively for knee arthroplasty to evaluate the patient's knee pain and functional status. Five studies [23, 39, 40, 52, 57] provided postoperative OKS for patients with no significant differences in baseline OKS levels across all studies ($P > .05$) (Supplementary Table S9). A total of 4177 patients were evaluated for OKS, with 2370 in the ERAS group and 1807 in the control group. The results show no statistical difference in the comparison between the ERAS and control groups [WMD = 0.33 (−0.93, 1.60), $P = .61$; $I^2 = 97.8\%$] (Table 2).

Effect of enhanced recovery after surgery on range of motion

Four studies [18, 28, 39, 55] reported postoperative ROM for patients. A total of 758 patients were evaluated for ROM, with 380 in the ERAS group and 378 in the control group. According to the results, ROM in the ERAS group was substantially greater than in the control group [WMD = 6.65 (2.09, 11.21), $P = .004$; $I^2 = 93.3\%$] (Table 2).

Publication bias

To evaluate publication bias, funnel plots of LOS, complication rate, readmission rate, transfusion rate, and mortality were performed. The P-value of Egger's test is <.05 in the funnel plots of complications, suggesting that there may be publication bias. The P-value of the Egger test in the funnel plots of other results is more than 0.05, roughly distributed in an inverted funnel, indicating a smaller impact of publication bias on the results (Supplementary Fig. S3.1–3.5).

Discussion

To our knowledge, this is the largest meta-analysis to discuss the effect of ERAS on arthroplasty to date. We included 11 RCTs and 36 nonrandomized control studies, with a total of 76 971 participants; 29 702 received the ERAS regimen and 47 269 received the control perioperative care. The included studies in this meta-analysis were conducted from 2008 to 2023. After our data analysis, the outcomes demonstrated that ERAS could reduce the hospital LOS and transfusion rates without increasing postoperative complications or readmission rates.

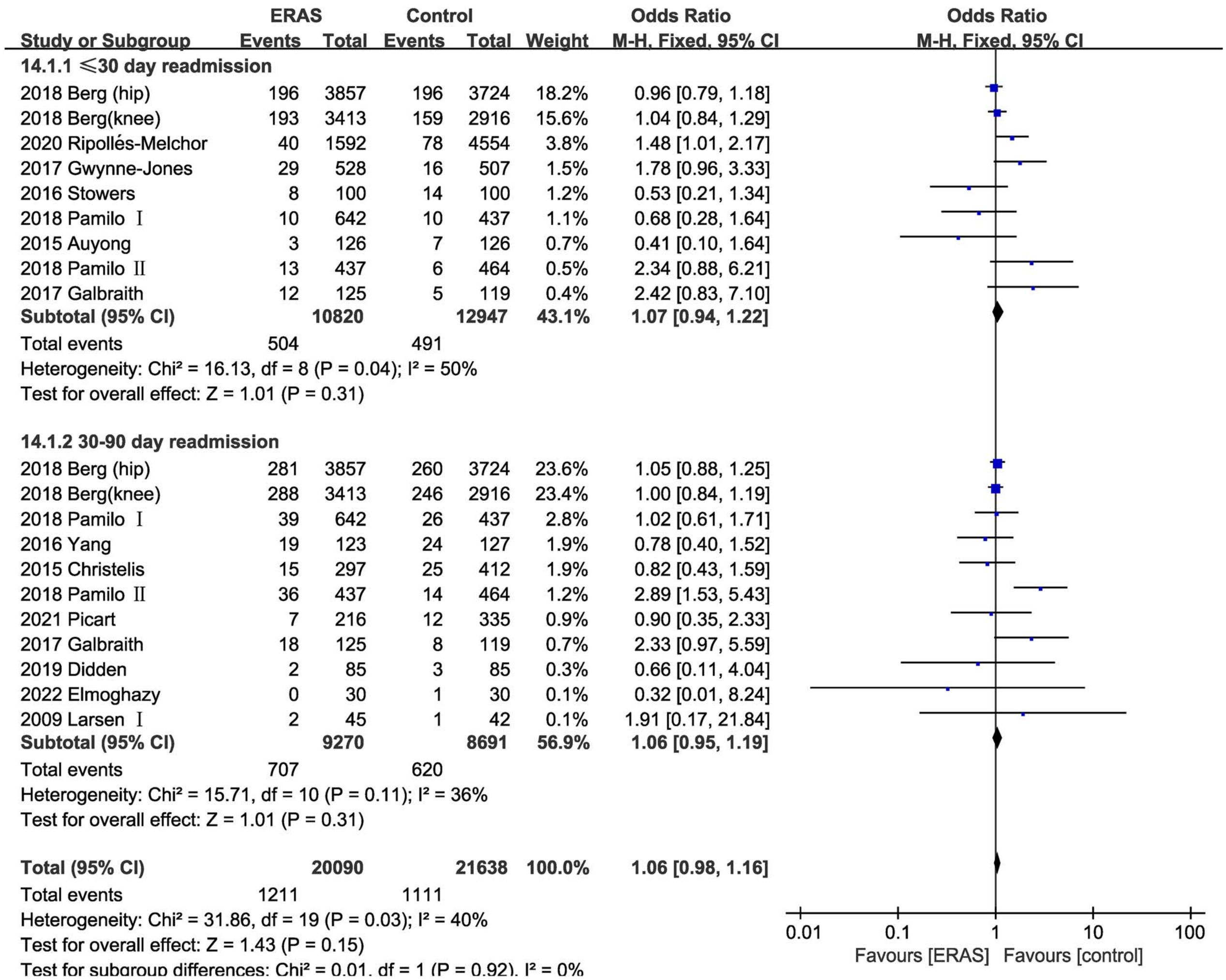

Figure 4 Comparison of readmission rate between the ERAS and control groups

ERAS promotes patients to be discharged from the hospital as soon as possible, saving societal medical resources, alleviating patients' financial burdens, and improving their quality of life. Although LOS heterogeneity was substantial, the direction and magnitude of effects [WMD = −2.65 (−3.03, −2.60), $P < .001$] show that ERAS protocols have important implications for LOS. Variations in the ERAS protocol, practice variation, diverse hospital resources and processes, and varying discharge criteria are considered the primary sources of substantial heterogeneity.

Considering that publishing bias is significant in the complication rate (Egger's test: $P = .002$), we employed the trim-and-fill method to evaluate the stability of the overall combined effect size. The findings revealed that the pooled OR outcome did not vary much ($P > .05$) (Supplementary Table S10), indicating that the postoperative complication rate was not significantly affected by publication bias and that the pooled OR outcome was essentially constant. Notably, although the ERAS group's incidence of postoperative complications was shown to be lower than the control group in the RCTs, the trim-and-fill method found no difference in the incidence of complications between patients in the two subgroups ($P > .05$) (Supplementary Table S10). In conclusion, ERAS protocols do not increase the incidence of complications in patients who underwent arthroplasty.

Moreover, ERAS does not increase the postoperative mortality rate. The subgroup analysis showed that ERAS could dramatically lower the 30-day mortality rate following surgery while not affecting the 30–90-day mortality rate. In addition, compared with the control group, the ERAS group had significantly lower VAS scores, better ROM, SF-36 BP, and SF-36 PF scores, and no differences in WOMAC and OKS. This indicates that ERAS may benefit patients who underwent arthroplasty by minimizing pain, increasing joint mobility, and enhancing the quality of life following surgery. In terms of lowering the demand for pricey narcotic medicines and improving patients' healing joint function, the ERAS group outperformed the control group.

During the analysis, we discovered that different articles' definitions of ERAS or fast-track protocols varied substantially. Due to this, we compiled a table of the ERAS guidelines' recommended elements and discovered that nearly all studies that used ERAS protocol for hip or knee arthroplasty advocated early mobilization. Even though many patients are discharged to rehabilitation centers or nursing homes for further rehabilitation, getting patients out of bed on the day of surgery helps them recover lower extremity function and cuts down on their hospital stay as well. The primary outcomes of the article were analyzed in subgroups based on the degree of compliance with highly advised ERAS elements recommended by previous reviews. In contrast to the low-degree literature, ERAS did not increase the readmission rate in the high-degree literature. However, the types of literature we classified as high-degree and low-degree did not differ regarding the LOS or the complication rate. This might be due to the fact that just a few research adopted certain recommendations from guidelines, such as preventing hypothermia, which could have affected how well our categorization of article quality was done. Additionally, it is possible that some researchers did not fully describe the ERAS used in their investigation, leading to the reduction of the reliability of our subgroup analysis. However, the clinical significance of our study of ERAS subgroups is still quite vital. To not increase the readmission rate of patients with THA or TKA, we hope that future research will use the seven ERAS elements suggested in our meta-analysis.

As an update and addition to earlier reviews [58–62], we conducted a rigorous literature review and used a sizeable sample in our meta-analysis. The ERAS approach and the conventional pathway for hip or knee arthroplasty were compared in every study. There are no language restrictions on studies, and studies were from various environments and geographic places. We also assessed 11 primary and secondary outcomes, including LOS, complications, readmission, transfusion, mortality, VAS, WOMAC, SF-36 BP, SF-36 PF, OKS, and ROM, which provided comprehensive assessment of ERAS. Additionally, our findings were strengthened by a thorough subgroup analysis based on the study type, surgical site, and degree of compliance with key ERAS elements.

Our research also had some limitations. There was a chance of bias in our meta-analysis because the authors of several RCTs did not fully specify allocation concealment or blinding procedures, which would have reduced the statistical precision. The included studies did not apply standardized and consistent clinical procedures for ERAS programs, leading to bias in the meta-analysis. Moreover, most pertinent clinical outcomes were evaluated; however, we did not evaluate expenditures, primarily due to the relative absence of these data in the arthroplasty group. Additionally, the test effectiveness of the results was decreased due to the small sample sizes for some of the secondary outcomes.

Conclusion

In THA and TKA patients, ERAS could significantly shorten the LOS, reduce transfusion rate, and lower 30-day postoperative mortality without increasing postoperative complications or readmission rate. Also, ERAS could decrease the VAS of patients while improving their ROM, SF-36 BP, and SF-36 PF scores. Finally, based on our findings, ERAS is better for patients' readmission rate in high-degree literature. These findings could support using these ERAS components in THA and TKA. However, further studies are needed to standardize these multimodal care pathways and better characterize these elements' impact on patients undergoing arthroplasty.

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Supplementary data

Supplementary data are available at *POSTMJ Journal* online.

Conflict of interest statement

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Data availability of data and material

All data generated or analyzed in this study are included in this published article.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Authors' contributions

Qingqing Zhang, designed study, collected data, analyzed data, wrote article; Yuzhang Chen, collected data, analyzed data, wrote article; Yi Li, collected data, wrote article; Ruikang Liu, designed study; Saroj Rai, designed study; Jin Li, designed study, analyzed data, revised article; Pan Hong, designed study, analyzed data, revised article.

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